

Probability*

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1. Pre-requisites

1.1 Bootcamp: Set

Set is a collection of objects. Members of set are called elements.

Notation:

For sets, $A, B, C, \dots$

For elements, $a, b, c, \dots$

For membership, $\in$ e.g. $a \in A$

For non membership, $\notin$.

For universal set, $\mathbb{U}$ i.e. everything.

For null set, ϕ .

Example:

$B = \{x/0 \leq x \leq 1\}$ where $/$ means such that.

$C = \{x/x \in \mathbf{R}, x^2 = -1\} = \phi$

Definition: If every element of A is an element of B then A is subset of B . i.e. $A \subset B$.

Definition: $A = B$ iff (if and only if) $A \subset B$ and $B \subset A$.

Properties:

- $\phi \subset A$; $A \subset U$; $A \subset A$
- $A \subset B, B \subset C \implies A \subset C$

Remark: The order in which the elements of set are listed is immaterial. E.g. $\{a, b, c\} = \{b, c, a\}$.

Definition: The complement of A with respect to U is $A^c = \{x \mid x \in U \text{ and } x \notin A\}$.

Definition: The intersection of A and B is $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.

Definition: The union of A and B is $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.

If $A \cap B = \phi$, then A and B are **disjoint** or **mutually exclusive**.

Definition:

- Minus: $A - B = A \cap B^c$
- Symmetric difference or XOR: $A \triangle B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

- The **cardinality** of A , denoted by $|A|$ is the number of elements in A . A is finite if $|A| < \infty$.
 $B = \{1, 2, 3, \dots\}$ is **countably infinite** i.e. $|B| = \aleph_0$
 $C = \{x | x \in [0, 1]\}$ is **uncountably infinite** i.e. $|C| = \aleph_1$

Laws of Operation:

- **Complement Law:** $A \cup A^c = U$, $A \cap A^c = \phi$, $(A^c)^c = A$
- **Commutative Law:** $A \cup B = B \cup A$, $A \cap B = B \cap A$
- **Associative Law:** $A \cup (B \cup C) = (A \cup B) \cup C$, $A \cap (B \cap C) = (A \cap B) \cap C$
- **Distributive Law:** $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- **De-Morgan's Law:** $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$

1.2 Bootcamp: Derivative

Definition: The function $f(x)$ maps values of X from a certain domain X to a certain range Y which can be denoted $f : x \rightarrow Y$.

If $f(x) = x^2$ then the function takes x -values from the real line $\mathbb{R}$ to the non-negative portion of real line $\mathbb{R}^+$.

Definition: We say that $f(x)$ is **continuous** function if for any x_0 & $x \in X$, we have $\lim_{x \rightarrow 0} f(x) = f(x_0)$ where $f(x)$ is assumed to exist for all $x \in X$.

The function $f(x) = 3x^2$ is continuous for all x . The function $f(x) = \lfloor x \rfloor$ i.e. round down to nearest integer e.g. $\lfloor 3.4 \rfloor = 3$. This function has discontinuity at any integer x .

Definition: The **inverse** of function $f : X \rightarrow Y$ is reverse mapping of $g : Y \rightarrow X$ such that $f(x) = y$ iff $g(y) = x$ for all appropriate x and y . The inverse is often written as f^{-1} and is especially useful if $f(x)$ strictly increasing or decreasing function. Note that $f^{-1}(f(x)) = x$.

Definition: If $f(x)$ is continuous, then it is **differentiable** if,

$$\frac{d}{dx} f(x) = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists and is well defined for given x . The derivative of $f(x)$ is slope of the function.

$$[x^k]' = kx^{k-1}$$

$$[e^x]' = e^x$$

$$[\sin(x)]' = \cos(x)$$

$$[\cos(x)]' = -\sin(x)$$

$$[\ln(x)]' = \frac{1}{x}$$

$$[\arctan(x)]' = \frac{1}{1+x^2}$$

Theorem: Some properties of derivatives

$$[af(x) + b]' = af'(x)$$

$$[f(x) + g(x)]' = f'(x) + g'(x)$$

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$$

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)}$$

$$[f(g(x))]' = f'(g(x))g'(x)$$

Remark: The second derivative $f''(x) = \frac{d}{dx}f'(x)$ and is the “slope of slope”. If $f(x)$ is position, then $f'(x)$ can be regarded as “velocity” and $f''(x)$ as “acceleration”.

The minimum or maximum of $f(x)$ can only occur when slope of $f(x)$ is 0, i.e. only when $f'(x) = 0$, say at the critical point $x = x_0$. Exception: Check the endpoints of your intervals of interest as well.

If $f''(x) < 0$, you get maximum, if $f''(x) > 0$, you get a minimum. If $f''(x) = 0$, you get a **point of inflection**.

1.3 Bootcamp: Integration

Definition: The function $F(x)$ having derivative $f(x)$ is called the **anti-derivative** or **indefinite integral**. It is denoted by $F(x) = \int f(x)dx$.

Fundamental Theorem of Calculus: If $f(x)$ is continuous, then the area under the curve for $x \in [a, b]$ is denoted and given by the **definite integral**.

$$\int_a^b f(x)dx = F(x)|_a^b = F(b) - F(a)$$

$$\int x^k dx = \frac{x^{k+1}}{k+1} + c \quad \text{for } k \neq -1 \text{ where } c \text{ is arbitrary constant}$$

$$\int \frac{dx}{x} = \ln|x| + c$$

$$\int e^x dx = e^x + c$$

$$\int \cos(x)dx = \sin(x) + c$$

$$\int \frac{1}{1+x^2} dx = \arctan(x) + c$$

Theorem: Some well known properties of definite integrals

$$\int_a^a f(x)dx = 0$$

$$\int_a^b f(x)dx = - \int_b^a f(x)dx$$

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

Theorem: Some other properties of general integrals:

$$\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx$$

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx \quad \text{integration by parts}$$

$$\int f(g(x))g'(x)dx = \int f(u)du \quad \text{Substitution rule with } u = g(x)$$

Definition: Derivative of arbitrary order K can be written as $f^k(x)$ or $\frac{d^k}{dx^k}f(x)$. By convention $f^0(x) = f(x)$.

The **Taylor Series Expansion** of $f(x)$ about a point a is given by

$$f(x) = \sum_{k=0}^{\infty} \frac{f^k(a)(x-a)^k}{k!}$$

The **Maclaurin Series** is simply Taylor expanded around $a = 0$.

Some famous **Maclaurin Series**;

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Here are some miscellaneous sums:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=0}^{\infty} \frac{1}{1-p} \quad (\text{for } -1 < p < 1)$$

Theorem: Occasionally, we run into trouble when taking indeterminate ratios of form $\frac{0}{0}$ or $\frac{\infty}{\infty}$. In such cases, **L' Hospital Rule** is useful. If the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both go to 0 or both go to ∞ , then,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Example:

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = \lim_{x \rightarrow 0} \frac{\cos(x)}{1} = 1$$

Double Integration:

Whereas single integrals get us the area under a curve, double integrals represent the volume under a three dimensional function.

The volume under $f(x, y) = 8xy$ over region $0 < x < y < 1$ is given by

$$\int_0^1 \int_0^y f(x, y) dx dy = \int_0^1 \int_0^y 8xy dx dy = \int_0^1 4y^3 dy = 1$$

We can swap the order of integration to get same answer.

$$\int_0^1 \int_x^1 8xy dy dx = \int_0^1 4x(1 - x^2) dx = 1$$

2. Introduction to Probability

2.1 Introduction

Mathematical models are either

- Deterministic (no uncertainty/randomness)
- Probabilistic (have some uncertainty)

Q. A couple has two kids and at least one is boy. What is the probability that both are boys?

Possibilities: GG, BG, GB, BB. Eliminate GG since we know that there's at least one boy. Then $P(BB) = \frac{1}{3}$.

Probability is methodology that describes the random variation in systems. **Statistics** uses data (sample) to draw conclusion about population.

Definition: A **sample space** associated with an experiment E is the set of all possible outcome of E. It's usually denoted by S or Ω .

Coin Toss: $S = \{H, T\}$

Toss a coin 2 times: $S : \{HH, HT, TH, TT\}$

Definition: An **event** is a set of possible outcomes. Thus, any subset of S is event.

Toss a dice, $S = \{1, 2, \dots\}$

If A is event "odd number occurs", $A = \{1, 3, 5\}$

The **empty set** ϕ is an event of S .

A is an event of S .

If A is an event, then A^c is the **complementary** event.

If A and B are events, then $A \cup B$ and $A \cap B$ are events.

Definition: The **Probability** of a generic event $A \subset S$ is a function that adheres to following axioms:

- $0 \leq P(A) \leq 1$ (probabilities are always between 0 and 1)
- $P(S) = 1$ (probability of some outcome is 1)
- If A and B are disjoint events, i.e. $A \cap B = \phi$ then, $P(A \cup B) = P(A) + P(B)$.
- Suppose $A_1, A_2, \dots$ is a sequence of disjoint events, i.e. $A_i \cap A_j = \phi$ for $i \neq j$.

$$\begin{aligned}
P(S) &= P(U_{i=1}^{\infty} A_i) \\
&= \sum_{i=1}^{\infty} P(A_i) \\
&= \sum_{i=1}^{\infty} \frac{1}{2^i}
\end{aligned}$$

Theorem: $P(A^c) = 1 - P(A)$

Proof:

$$\begin{aligned}
1 &= P(S) \\
&= P(A \cup A^c) \\
&= P(A) + P(A^c) \quad \because A \cap A^c = \phi
\end{aligned}$$

Corollary: $P(\phi) = 0$

Proof: By definition, $\phi = S^c$; so the result follows the theorem and axiom 2. **Remark:** The converse is false: $P(A) = 0$ doesn't imply $A = \phi$.

Theorem: For any two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof: Use Venn-diagram.

Remark: Axiom 3 is special case of this theorem with $A \cap B = \phi$.

Theorem: For any three events A , B and C ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Theorem: Here is the **Principle of inclusion-exclusion**:

$$\begin{aligned}
P(A_1 \cup A_2 \cup \dots \cup A_n) &= \sum_{i=1}^n P(A_i) - \sum \sum_{i < j} P(A_i \cap A_j) + \sum \sum \sum_{i < j < k} P(A_i \cap A_j \cap A_k) \\
&\quad + \dots + (-1)^{n-1} P(A_1 \cap A_2 \cap \dots \cap A_n)
\end{aligned}$$

Remark: You “include” all of the “single” events, “exclude” the double events, include the triple events etc.

Finite Sample Space:

Suppose S is finite $S = S_1, S_2, \dots, S_n$. Finite sample space often allows us to calculate the probabilities of certain events more efficiently. To illustrate, let $A \subset S$ be any event, then $P(A) =$

$$\Sigma_{S_i \in A} P(S_i).$$

You have 2 red cards, a blue and a yellow card. Pick a card at random then,

$$S = \{S_1, S_2, S_3\} = \{red, blue, yellow\}$$

$$P(S_1) = \frac{1}{2} \quad P(S_2) = \frac{1}{4} \quad P(S_3) = \frac{1}{4}$$

$$P(\text{red or yellow}) = \frac{1}{2} + \frac{1}{4}$$

Definition: A **simple sample space (SSS)** is a finite sample space in which outcomes are equally likely.

Remark: In above example, S is not simple sample space since $P(S_1) \neq P(S_2)$.

Example: Toss 2 fair coins,

$S = \{HH, HT, TH, TT\}$ is a *SSS* (all probabilities are $\frac{1}{4}$).

Theorem: For any event A in *SSS*,

$$P(A) = \frac{|A|}{|S|} = \frac{\text{no. of elements in } A}{\text{no. of elements in } S}$$

2.1.1 Counting Techniques

Muffin (blueberry or oatmeal) or a bagel (sesame, plain, salt, garlic) but not both. You have $2 + 4 = 6$ choices in total.

$n_{AB} = 3$ ways to go from city A to B (walk, car, bus) and $n_{BC} = 4$ ways to go from B to C (car, bus, train, plane). Then you can go from A to C (via B) using $n_{AB} \cdot n_{BC} = 3 * 4 = 12$ ways.

Roll two dice. How many outcomes?

$(3, 2) \neq (2, 3)$ so, answer $= 6 * 6 = 36$ ways.

Toss n dice. Outcome $= 6^n$ possibilities.

Toss n coins. Outcome $= 2^n$ possibilities.

2.1.2 Permutation

An arrangement of n symbols in a **definite order** is a **permutation** of n symbols.

Example: How many ways to arrange 1, 2, 3 ?

Answer: 6 ways: 123, 132, 213, 312, 321, 231

- **Number of ways to arrange 1, 2, ..., $n = n * (n - 1) * (n - 2) * \dots * 2 * 1 = n!$

Definition: The number of **r-tuples** we can make from n different symbols (each used at most once) is called the **number of permutations of n things taken r at a time**.

$$P_{n,r} = \frac{n!}{(n-r)!}$$

Note: $0! = 1$ & $P_{n,n} = n!$

Proof:

$$\begin{aligned} P_{n,r} &= (\text{choose first})(\text{choose second})\dots(\text{choose } r^{\text{th}}) \\ &= n(n-1)(n-2)\dots(n-r+1) \\ &= \frac{n(n-1)\dots(n-r+1)(n-r)\dots 2 * 1}{(n-r)\dots 2 * 1} \\ &= \frac{n!}{(n-r)!} \end{aligned}$$

Example: How many license plates of 6 digits can be formed from numbers $\{1,2,\dots,9\}$? + with no repetitions: $P_{9,3} = 60480$ + with repetitions: $9 * \dots * 9 = 9^6$ ways + containing repetitions: $9^6 - 60480 = 470961$

2.1.3 Combination

How many subsets of $\{1, 2, 3\}$ contain exactly 2 elements? (order isn't important)

Answer: 3 subsets - $\{1, 2\}, \{1, 3\}, \{2, 3\}$

Definition: The number of subsets with r elements of a set with n elements is called **number of combinations of n things taken r at a time**.

Notation: $C_{n,r}$ or $\binom{n}{r}$. These are also called **binomial coefficients**.

$$C_{n,r} = \frac{n!}{r!(n-r)!}$$

The difference between permutation and combination:

- Combination: $(a, b, c) = (b, a, c)$ i.e. order doesn't concern,
- Permutation: $(a, b, c) \neq (b, a, c)$ i.e. concerned with order.

Choosing a permutation is same as first choosing a combination and putting the elements in order.

$$\frac{n!}{(n-r)!} = \binom{n}{r} r!$$

$$\frac{n!}{(n-r)!r!} = \binom{n}{r}$$

Following results should be intuitive:

- $\binom{n}{r} = \binom{n}{n-r}$
- $\binom{n}{0} = \binom{n}{n} = 1$
- $\binom{n}{1} = \binom{n}{n-1} = n$

2.1.4 Binomial Theorem

$$(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^i y^{n-i}$$

This is where **Pascal's triangle** comes from.

Corollary: Surprising fact

$$\sum_{i=0}^n \binom{n}{i} = 2^n$$

Proof: By the binomial theorem:

$$2^n = (1 + 1)^n$$

$$= \sum_{i=0}^n \binom{n}{i} 1^i 1^{n-i}$$

2.1.5 Problems

Q. Select 2 cards from a deck without replacement and care about order? Possibilities
 $= 52 * 51 = 2652$ ways.

Q. Box of 10 sox - 2 red and 8 black. Pick 2 without replacement.

- Let A be event that both are red.

$$P(A) = \frac{\text{ways to pick 2 reds}}{\text{ways to pick 2 sox}} = \frac{2*1}{10*9} = \frac{1}{45}$$

- Let B be event that both are black.

$$P(B) = \frac{8*9}{10*9} = \frac{28}{45}$$

- Let C be one of each color. Since, A and B are disjoint,

$$P(C) = 1 - P(C^c) = 1 - P(A \cup B) = 1 - \frac{1}{45} - \frac{28}{45} = \frac{16}{45}$$

Q. An NBA team has 12 players. How many ways can the coach choose the starting 5?

$$\binom{12}{5} = \frac{12!}{5!7!} = 792$$

Q. Smith is one of the players on the team. How many of 792 starting lineup include him?

$$\binom{11}{4} = \frac{11!}{4!7!} = 330$$

Q. 4 red marbles, 2 whites. Put them in random order.

a. $P(2 \text{ end marbles are W})$

$S = \{\text{Possible pairs of slots that W's occupy}\}$

$$|S| = \binom{6}{2} = \frac{6!}{2!(6-2)!} = 15$$

Since, W's must occupy end slots so, $|A| = \binom{2}{2} = 1$

$$P(A) = \frac{|A|}{|S|} = \frac{1}{15}$$

b. $P(2 \text{ end marbles aren't both W}) = 1 - P(A) = \frac{14}{15}$

c. $P(2 \text{ W's are side by side})$

WWRRRR or RWWRRR or RRWWRR or RRRWWRR or RRRRWW

$$|B| = 5$$

$$P(B) = \frac{5}{15}$$

2.2 Hypergeometric Distribution

Definition: You have a objects of type 1 and b objects of type 2. Select n objects **without replacement** from $a + b$ objects. Then,

$$\begin{aligned} P(\text{k type 1's were picked}) &= \frac{(\text{Number of ways to choose k type 1's out of a})(\text{Choose n-k type 2's out of b})}{(\text{Number of ways to choose n out of a+b})} \\ &= \frac{\binom{a}{k} \binom{b}{n-k}}{\binom{a+b}{n}} \end{aligned}$$

The number of type 1's chosen is said to have the **hypergeometric distribution**.

Example: 3 sox in box with $a = 2$ red, $b = 1$ blue. Pick $n = 3$ without replacement.

$$\begin{aligned} P(\text{Exactly k=2 reds are picked}) &= \frac{\binom{a}{k} \binom{b}{n-k}}{\binom{a+b}{n}} \\ &= \frac{\binom{2}{2} \binom{1}{1}}{\binom{3}{3}} \\ &= 1 \end{aligned}$$

2.3 Binomial Distribution

Definition: You again have a objects of type 1 and b objects of type 2. Now, select n objects **with replacement** from $a + b$ objects.

$$P(\text{k type 1's were picked}) = (\text{Number of ways to choose k 1's and n-k 2's})$$

$$P(\text{Choose k 1's in a row then n-k 2's in a row})$$

$$P(\text{k type 1's were picked}) = \binom{n}{k} \left(\frac{a}{a+b} \right)^k \left(\frac{b}{a+b} \right)^{n-k}$$

2.4 Multinomial Coefficients

Example: n_1 blue sox, n_2 reds. The number of assortments is $\binom{n_1+n_2}{n_1}$. Generalization for k types of objects: $n = \sum_{i=1}^k n_i$ The number of arrangements is

$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$$

This is known as **multinomial coefficient**.

Example: How many ways letters in “MISSISSIPPI” be arranged?

$$\frac{\text{Number of permutations of 11 letters}}{(\text{Number of M's})(\text{Number of P's})(\text{Number of I's})(\text{Number of S's})}$$

$$= \frac{11!}{1!2!4!4!}$$

2.5 Conditional Probability

The probability of A occurs given B occurs is

$$P(A/B) = \frac{|A \cap B|}{|B|} = \frac{\frac{|A \cap B|}{|S|}}{\frac{|B|}{|S|}} = \frac{P(A \cap B)}{P(B)}$$

Definition: If $P(B) > 0$, the conditional probability of A given B is

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Remark: If A and B are disjoint, then $P(A/B) = 0$. If B occurs, there is no chance that A can occur.

What happens if $P(B) = 0$? In that case, no need to consider $P(A/B)$.

Example: Toss 2 dice and take the sum.

A: odd toss = {3, 5, 7, 9, 11}

B: {2, 3}

$$P(A) = P(3) + \dots + P(11) = \frac{2}{36} + \frac{4}{36} + \dots + \frac{2}{36} = \frac{1}{2}$$

$$P(B) = \frac{1}{36} + \frac{2}{36} = \frac{1}{12}$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{36}}{\frac{1}{12}} = \frac{2}{3}$$

Example: A couple has two kids and at least one is boy. What's the probability that both are boys?

$S = \{GG, GB, BG, BB\}$

C : Both are boys = {BB}

D : At least 1 boy = {GB, BG, BB}

$$P(C/D) = \frac{P(C \cap D)}{P(D)} = \frac{P(C)}{P(D)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Example: A couple has two kids and at least one is born on tuesday. What is the probability that both are boys?

$$B_x[G_x] = Boy[Girl]$$

born on day x; $x = 1, 2, \dots, 7$

$x = 3$ is Tuesday.

$$S = \{(G_x, G_y), (G_x, B_y), (B_x, G_y), (B_x, B_y), x, y = 1, 2, \dots, 7\}$$

$$\text{So, } |S| = 4 * 49 = 196$$

i.e. 4 combination of B and G and 49 combination of x and y.

C: Both are boys (with at least one born on tuesday)

$$= \{(B_x, B_3), x = 1, 2, \dots, 7\} \cup \{(B_3, B_y), y = 1, 2, \dots, 7\}$$

Note: $|C| = 13$ {to avoid double counting (B_3, B_3) }

D: There is at least one boy born on Tuesday.

$$= C \cup \{(G_x, B_3), (B_3, G_y), x, y = 1, 2, \dots, 7\}$$

$$|D| = 27$$

$$P(C/D) = \frac{P(C \cap D)}{P(D)} = \frac{P(C)}{P(D)} = \frac{\frac{13}{196}}{\frac{27}{197}} = \frac{13}{27}$$

Properties: Analogous to axioms of probability

- $0 \leq P(A/B) \leq 1$
- $P(S/B) = 1$
- $A_1 \cap A_2 = \phi \rightarrow P(A_1 \cap A_2/B) = P(A_1/B) + P(A_2/B)$
- If $A_1, A_2, \dots$ are all disjoint then

$$P(\bigcup_{i=1}^{\infty} A_i/B) = \sum_{i=1}^{\infty} P(A_i/B)$$

2.6 Independence

Any unrelated events are independent.

Example:

A: It rains on Mars tomorrow.

B: Coin lands on H.

Definition: A & B are independent iff $P(A \cap B) = P(A).P(B)$

Remark: If $P(A) = 0$, then A is independent of any other event.

Remark: Events don't have to be physically unrelated to be independent.

Theorem: Suppose $P(B) > 0$. Then A and B are independent $\leftrightarrow P(A/B) = P(A)$.

Proof: A & B independent $\leftrightarrow P(A \cap B) = P(A).P(B) \leftrightarrow \frac{P(A \cap B)}{P(B)} = P(A)$

Remark: So, if A and B are independent, the probability of A doesn't depend on whether or not B occurs.

Bayes Theorem: A and B are independent $\leftrightarrow A'$ and B' are also independent.

Proof: Only need to prove in $\rightarrow$ direction (then $\leftarrow$ follows trivially).

$$P(A) = P(A \cap B') + P(A \cap B)$$

So,

$$\begin{aligned} P(A \cap B') &= P(A) - P(A \cap B) \\ &= P(A) - P(A).P(B) \quad \{A, B \text{ are independent}\} \\ &= P(A)\{1 - P(B)\} \\ &= P(A).P(B') \end{aligned}$$

Don't confuse independence with disjointness!

Theorem: If $P(A) > 0$ and $P(B) > 0$, A and B can't be independent and disjoint at the same time.

Proof: Suppose A and B are disjoint, $A \cap B = \phi$. Then, $P(A \cap B) = 0 < P(A).P(B)$. Thus, A and B aren't independent. Similarly, independent doesn't imply disjoint.

Remark: In fact, independence and disjointness are almost opposite. If A and B are disjoint and A occurs, then you have information that B cannot occur. So, A and B can't be independent.

Extension to more than two events:

Definition: A, B, C are independent iff

- $P(A \cap B \cap C) = P(A).P(B).P(C)$

- All pairs are independent:

$$P(A \cap B) = P(A).P(B)$$

$$P(A \cap C) = P(A).P(C)$$

$$P(B \cap C) = P(B).P(C)$$

General Definition: $A_1, \dots, A_k$ are independent iff $P(A_1 \cap \dots \cap A_k) = P(A_k)$ and all subsets of $\{A_1, \dots, A_k\}$ are independent.

Independent Trials: Perform n trials of an experiment such that the outcome of one trial is independent of outcomes of other trials. Eg. Flip 3 coins independently.

Remark: For independent trials, you just multiply the individual probabilities.

Eg. Flip a coin infinitely many times (each flip is independent of others).

$$\begin{aligned}
 P_n &= P(\text{First H on } n\text{th trial}) \\
 &= P(\underbrace{TT\dots T}_{n-1} H) \\
 &= \underbrace{P(T).P(T)\dots P(T)}_{n-1}.P(H) \\
 &= \left(\frac{1}{2}\right)^{n-1} \cdot \frac{1}{2} = \frac{1}{2^n} \\
 &= \frac{1}{2^n} \quad \{\text{Each has probability } 1/2\} \\
 P(H \text{ eventually}) &= \sum_{n=1}^{\infty} P_n \\
 &= \sum_{n=1}^{\infty} 2^{-n} \\
 &= 1
 \end{aligned}$$

2.7 Partitions and laws of probability

Partition of Sample Space split the sample space into disjoint, yet all encompassing subsets.

Definition: The events $A_1, A_2, \dots, A_n$ form a partition of sample space S if

- $A_1, A_2, \dots, A_n$ are disjoint.
- $\bigcup_{i=1}^n A_i = S$
- $P(A_i) > 0$ for all i .

Remark: When an experiment is performed, exactly one A_i 's occur.

Example: A and A' form partition.

Suppose $A_1, A_2, \dots, A_n$ form partition of S and B is arbitrary event. Then,

$$B = \bigcup_{i=1}^n (A_i \cap B)$$

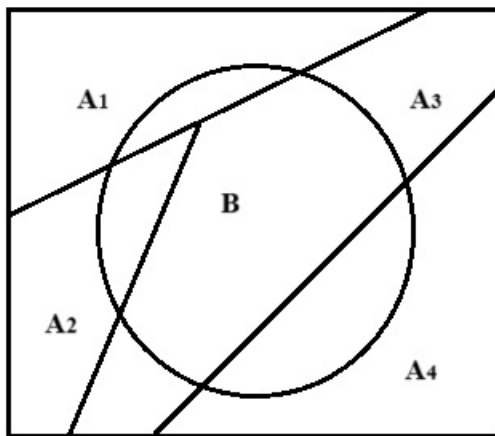


Figure 1: Partitions

$$\begin{aligned} P(B) &= P[\bigcup_{i=1}^n (A_i \cap B)] \\ &= \sum_{i=1}^n P(A_i \cap B) \quad (\text{Since } A_1, A_2, \dots, A_n \text{ are disjoint}) \\ &= \sum_{i=1}^n P(A_i)P(B/A_i) \quad (\text{Definition of conditional Probability}) \end{aligned}$$

This is **law of probability**.

Example: Suppose we have 10 Georgia Tech students and 20 University of Georgia students taking a test. GT students have 95% chance of passing but UGA have 50%. Determine probability that he/she passes.

$$P(\text{passes}) = P(GT)P(\text{passes}/GT) + P(UGA)P(\text{passes}/UGA)$$

2.8 Bayes Theorem

Immediate consequence of law of total probability.

Bayes Theorem: If $A_1, A_2, \dots, A_n$ form partition of S and B is any event then,

$$\begin{aligned} P(A_j/B) &= \frac{P(A_j \cap B)}{P(B)} \\ &= \frac{P(A_j)P(B/A_j)}{\sum_{i=1}^n P(B/A_i)} \end{aligned}$$

The $P(A_j)$'s are prior probabilities ("before B").

The $P(A_j/B)$'s are posterior probabilities ("after B").

The $P(A_j/B)$'s add up to 1.

2.9 Probability Problems

Birthday Problem

Q. There are n people in room. Find the probability that at least two have the same birthday. (Ignore Feb 29 and assume that all 365 days have equal probability.

The (simple) sample size is $S = \{(x_1, \dots, x_n) : x_i \in \{1, 2, \dots, 365\}, V_i\}$

(x_i is person i 's birthday) and note that $|S| = (365)^n$.

Let A: All birthdays are different then,

$$\begin{aligned} P(A) &= \frac{(365)(364)\dots(365 - n + 1)}{365^n} \\ &= 1 \cdot \frac{364}{365} \cdot \frac{363}{365} \dots \frac{365 - n + 1}{365} \end{aligned}$$

$$P(A') = 1 - P(A)$$

When, $n = 366$, $P(A') = 1$

For, $P(A') > \frac{1}{2}$, n must be ≥ 23 .

When, $n = 50$, $P(A') = 0.97$, $P(A')$ is probability of at least one birthday match (not unique).

The Envelope Problem

Q. A group of n people receives n envelopes with their name on them but someone has completely mixed up the envelopes. Find the probability that at least one person will receive the proper envelope.

Let A_i : Person i receive correct envelope.

We want $P(A_1 \cup A_2 \dots \cup A_n)$

By principle of Inclusion-Exclusion,

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i) - \sum \sum_{i < j} P(A_i \cap A_j) + \sum \sum \sum_{i < j < k} P(A_i \cap A_j \cap A_k) \\ + \dots + (-1)^{n-1} P(A_1 \cap A_2 \cap \dots \cap A_n)$$

Since all $P(A_i)$'s are same, all of $P(A_i \cap A_j)$'s are the same.

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = nP(A_1) - \binom{n}{2} P(A_1 \cap A_2) + \binom{n}{3} P(A_1 \cap A_2 \cap A_3) + \dots + (-1)^{n-1} P(A_1 \cap A_2 \dots \cap A_n)$$

$$\begin{aligned} P(A_1) &= \frac{1}{n} \\ P(A_2) &= \frac{1}{n-1} \\ P(A_1 \cap A_2) &= \frac{1}{n(n-1)} \\ P(A_1 \cup A_2 \cup \dots \cup A_n) &= \frac{n}{n} - \binom{n}{2} \frac{1}{n} \cdot \frac{1}{n-1} + \binom{n}{3} \frac{1}{n} \cdot \frac{1}{n-1} \cdot \frac{1}{n-2} + \dots + (-1)^{n-1} \frac{1}{n!} \\ &= 1 - \frac{1}{2!} + \frac{1}{3!} + \dots + (-1)^{n-1} \frac{1}{n!} \\ &= 1 - \frac{1}{e} \quad \{\text{Very similar to Mclaurin Series}\} \\ &= 0.6321 \end{aligned}$$

If $n = 4$ envelopes:

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3 \cup A_4) &= 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} \\ &= 0.625 \end{aligned}$$

3. Random Variables

3.1 Univariate Random Variables

Definition: A **random variable (RV)** is a function from the sample space to the real line. $X : S \rightarrow \mathbb{R}$.

Example: Flip 2 coins: $S = \{HH, HT, TH, TT\}$

Suppose X is RV corresponding to the number of H 's,

$$X(TT) = 0, X(HT) = 1, X(HH) = 2$$

$$P(X = 0) = \frac{1}{4}, P(X = 1) = \frac{1}{2}, P(X = 2) = \frac{1}{4}$$

Notation: Capital letters like X, Y, Z usually represent RV's. Small letters like x, y, z represent particular values of RV's.

Example: Flip a coin

$$X = \begin{cases} 1 & \text{if T} \\ 0 & \text{if H} \end{cases}$$

Roll a die

$$Y = \begin{cases} 0 & \text{if } \{1,2,3\} \\ 1 & \text{if } \{4,5,6\} \end{cases}$$

For our purpose, X and Y are same, since $P(X = 0) = P(Y = 0) = \frac{1}{2}$ and $P(X = 1) = P(Y = 1) = \frac{1}{2}$.

Example: Select a real number at random between 0 and 1. There are infinite number of “equally likely” outcome.

Conclusion: $P(\text{we choose the individual point } x) = P(X = x) = 0$.

But $P(X \leq 0.65) = 0.65$ and $P(X \in [0.3, 0.7]) = 0.4$.

If A is an interval in $[0, 1]$ then $P(X \in A)$ is the length of A .

Definition: If a number of possible values of a RV X is finite or countably infinite then X is **discrete** RV otherwise,

A **continuous** RV is one with probability 0 at every point.

Example:

- Flip a coin - get H or T. Discrete
- Pick a point at random in $[0, 1]$. Continuous
- The amount of time you wait in line is either 0 (with positive probability) or some positive real number - a combined discrete - continuous RV.

3.1.1 Discrete Random Variable

Definition: If X is discrete RV, its **probability mass function (pmf)** is

$$f(x) = P(X = x)$$

Note that $0 \leq f(x) \leq 1$, $\sum_x f(x) = 1$

Example: Flip 2 coins. Let X be number of heads.

$$f(x) = \begin{cases} \frac{1}{4} & \text{if } x = 0 \text{ or } 2 \\ \frac{1}{2} & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

3.1.1.2 Uniform Distribution: Uniform distribution of integers $1, 2, \dots, n$. X can equal $1, 2, \dots, n$ each with probability $\frac{1}{n}$.

$$f(i) = \frac{1}{n} \quad i = 1, 2, \dots, n$$

3.1.1.2 Binomial Distribution: Let X denote number of “successes” from n independent trials such that P (success) at each trial is p ($0 \leq p \leq 1$). Then X has the binomial distribution with parameters n and p . The trials are referred to as **Bernoulli Trials**.

Notation: $X \sim \text{Bern}(n, p)$

Example: Roll a die 3 independent times. Find P (Get exactly two 6’s)

“success (6)” and “failure” (1,2,3,4,5)

All trials are independent, $P(\text{success}) = \frac{1}{6}$ doesn’t change from trial to trial.

Let X = number of 6’s. Then $X \sim \text{Bern}(3, \frac{1}{6})$.

Theorem: If $X \sim \text{Bern}(n, p)$ then probability of k successes in n trials is

$$P(X = k) = \binom{n}{k} p^k q^{n-k}$$

where,

$$K = 0, 1, \dots, n \quad \text{and} \quad q = 1 - p$$

Proof: Consider the particular sequence of success and failures.

$$\begin{array}{cc} \underline{SS\dots S} & \underline{FF\dots F} \\ k \text{ success} & n-k \text{ failure} \\ \text{prob} = p^k q^{n-k} \end{array}$$

The number of ways to arrange the sequence is $\binom{n}{k}$.

Example: Roll 2 dice and get the sum. Repeat 12 times. Find P(Sum will be 7 or 11 exactly 3 times).

$$\begin{aligned} P(7 \text{ or } 11) &= P(7) + P(11) \\ &= \frac{7}{36} + \frac{2}{36} \\ &= \frac{2}{9} \end{aligned}$$

So, $X \sim \text{Bin}(12, \frac{2}{9})$ then,

$$P(X = 3) = \binom{12}{3} \left(\frac{2}{9}\right)^3 \left(\frac{7}{9}\right)^9$$

3.1.1.3 Poisson Distribution: If $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, $k = 0, 1, 2, \dots, \lambda$, $\lambda > 0$, we say that X has the **Poisson distribution** with parameter λ .

Notation: $X \sim \text{Pois}(\lambda)$

Example: Suppose the number of raisins in a cup of cookie dough is $\text{Pois}(10)$. Find the probability that cup of dough has at least 4 raisins.

$$\begin{aligned} P(X \geq 4) &= 1 - P(X = 0, 1, 2, 3) \\ &= 1 - e^{-10} \left(\frac{10^0}{0!} + \frac{10^1}{1!} + \frac{10^2}{2!} + \frac{10^3}{3!} \right) \\ &= 0.9897 \end{aligned}$$

3.1.2 Continuous Random Variables

Example: Pick a point X randomly between 0 and 1 and define the continuous function.

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Intuitively, if $0 \leq a \leq b \leq 1$ then,

$$\begin{aligned} P(a \leq X \leq b) &= \text{the "area" under } f(x) \text{ from } a \text{ to } b \\ &= b - a \end{aligned}$$

Definition: Suppose X is a continuous RV, the magic function $f(x)$ is **probability density function (PDF)** if

- $\int_{\mathbb{R}} f(x)dx = 1$ (area under the $f(x)$ is 1)
- $f(x) \geq 0$
- If $A \subseteq \mathbb{R}$, then $P(X \in A) = \int_A f(x)dx$ (probability that X is in a certain region of A)

Remark: If X is continuous RV then,

$$P(a < X < b) = \int_a^b f(x)dx$$

An individual point has probability 0 i.e. $P(x = x) = 0$.

If X is discrete then $f(x) = P(X = x)$ and must have $0 \leq f(x) \leq 1$.

If X is continuous,

- $f(x)$ is continuous,
- Instead think of $f(x)dx \approx P(x < X < x + dx)$.
- Must have $f(x) \geq 0$ and possibly > 1 .

3.1.2.1 Uniform Distribution If X is “equally likely” to be anywhere between a and b then X has the uniform distribution on (a, b) .

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$$

Notation: $X \sim Unif(a, b)$

Remark: $\int_{\mathbb{R}} f(x)dx = \int_a^b \frac{1}{b-a}dx = 1$

Example: If $X \sim Unif(-2, 8)$ then,

$$P(-1 < X < 6) = \int_{-1}^6 \frac{1}{8 - (-2)}dx = 0.7$$

3.1.2.2 Exponential Distribution X has the exponential distribution with parameter $\lambda > 0$ if it has PDF $f(x) = \lambda e^{-\lambda x}$, for $x \geq 0$.

Notation: $X \sim Exp(\lambda)$

Remark: $\int_{\mathbb{R}} f(x)dx = \int_0^{\infty} \lambda e^{-\lambda x}dx = 1$

Example: Suppose X is a continuous RV with PDF

$$f(x) = \begin{cases} cx^2 & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

First of all, let's find c. Noting that PDF must integrate to 1 we have,

$$1 = \int_{\mathbb{R}} f(x) dx$$

$$1 = \int_0^2 cx^2 dx$$

$$1 = \left. \frac{cx^3}{3} \right|_0^2$$

$$1 = \frac{8c}{3}$$

$$c = \frac{3}{8}$$

this means

$$f(x) = \frac{3x^2}{8}$$

$$\begin{aligned} P(0 < X < 1 \mid \tfrac{1}{2} < X < \tfrac{3}{2}) &= \frac{P(0 < X < 1 \cap \tfrac{1}{2} < X < \tfrac{3}{2})}{P(\tfrac{1}{2} < X < \tfrac{3}{2})} \\ &= \frac{P(\tfrac{1}{2} < X < 1)}{P(\tfrac{1}{2} < X < \tfrac{3}{2})} \\ &= \frac{\int_{\frac{1}{2}}^1 \frac{3}{8} x^2 dx}{\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{3}{8} x^2 dx} \\ &= \frac{7}{26} \end{aligned}$$

X has the **standard normal distribution** if its PDF is

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for all } x \in \mathbb{R}$$

3.1.3 Cumulative Probability Distribution

Definition: For any RV (discrete or continuous), the cumulative distribution function (cdf) is defined for all x by,

$$F(x) = P(X \leq x)$$

For X discrete,

$$F(x) = \sum_{\{y/y \leq x\}} f(y) = \sum_{\{y/y \leq x\}} P(X = y)$$

For X continuous,

$$F(x) = \int_{-\infty}^x f(y) dy$$

Example: Flip a coin twice. let X = number of H's.

$$X = \begin{cases} 0 \text{ or } 2 \text{ with prob } \frac{1}{4} \\ 1 \text{ with prob } \frac{1}{2} \end{cases}$$

The CDF is a step function

$$F(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{4} & \text{if } 0 \leq x < 1 \\ \frac{3}{4} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

Explanation

X defined as number of heads from two independent flips of a fair coin. $X = 0$ (no heads), $X = 1$ (one head) and $X = 2$ (two heads). The probability distribution of X follows binomial distribution i.e. $X \sim \text{Binom}(2, \frac{1}{2})$.

$$P(X = 0) = \binom{2}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$P(X = 1) = \binom{2}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^1 = \frac{1}{2}$$

$$P(X = 2) = \binom{2}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^0 = \frac{1}{4}$$

For $x < 0$, $F(x) = 0$.

For $0 \leq x < 1$, $F(x) = P(X = 0) = \frac{1}{4}$.

For $1 \leq x < 2$, $F(x) = P(X = 0) + P(X = 1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$

For $x \geq 2$, $F(x) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$

Warning: For discrete RVs, be careful about $\leq$ vs $<$ as the endpoints of the range (where step function jumps).

Theorem (Continuous CDF): If X is continuous RV, then $f(x) = F'(x)$ (assuming the derivative exists.)

Proof:

$$F'(x) = \frac{d}{dx} \int_{-\infty}^x f(t)dt = f(x), \quad \text{by the fundamental theorem of calculus}$$

Example: $X \sim Unif(0, 1)$. The PDF and cdf are

$$f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } 0 < x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

Explanation

$$f(x) = \frac{1}{1-0} = 1 \quad \text{for } 0 < x < 1$$

So,

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$$

For $x \leq 0$, $F(x) = 0$ since all values are between 0 and 1.

For $0 < x < 1$, $F(x) = \int_{-\infty}^x 1dt = x$.

For $x \geq 1$, $F(x) = 1$ since $x \geq 1$ includes all the probability.

Example: $X \sim Exp(\lambda)$

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$F(x) = \int_{-\infty}^x f(t)dt = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - e^{-\lambda x} & \text{if } x > 0 \end{cases}$$

We can use CDF to find **median** of X that is the point m such that,

$$0.5 = F(m) = 1 - e^{-\lambda m} \implies m = \left(\frac{1}{\lambda}\right) \ln(2)$$

Explanation

For, $x \leq 0$, $f(x) = 0$. So, $F(x) = 0$.

For, $x > 0$, $F(x) = \int_0^x \lambda e^{-\lambda x} dx = 1 - e^{-\lambda x}$

Properties of CDF

$F(x)$ is non-decreasing in x i.e. $a < b$ implies $F(a) \leq F(b)$.

$$\lim_{x \rightarrow \infty} F(x) = 1 \quad \lim_{x \rightarrow -\infty} F(x) = 0$$

$F(x)$ is right continuous at every point x .

Theorem: $P(X > x) = 1 - F(x)$

Proof:

By complements,

$$P(X > x) = 1 - P(X \leq x) = 1 - F(x)$$

Theorem: $a < b \implies P(a < X \leq b) = F(b) - F(a)$.

Proof: Since $a < b$, we have,

$$\begin{aligned} P(a < X \leq b) &= P(X > a, X \leq b) \\ &= P(X > a) + P(X \leq b) - P(X > a \cup X \leq b) \\ &= 1 - F(a) + F(b) - 1 \\ &= F(b) - F(a) \end{aligned}$$

where,

$$P(X > a) = 1 - F(a) \quad P(X \leq b) = F(b) \quad P(X > a \cup X \leq b) = 1$$

3.1.4 Great Expectations

Definition: The **mean** or **expected value** or **average** of random variable X is

$$\mu = E(x) = \begin{cases} \sum_x x f(x) & \text{if } X \text{ is discrete} \\ \int_{\mathbb{R}} x f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

The mean gives an indication of RV's central tendency. Think of it as a weighted average of the possible x 's where the weights are given by $f(x)$.

Example: Suppose X has the Bernoulli distribution with parameter p i.e. ($P(X = 1) = p$ and $P(X = 0) = q = 1 - p$). Then,

$$E(x) = \sum_x x f(x) = 1.p + 0.q = p$$

Example: Die toss. $X = 1, 2, \dots, 6$ each with probability $\frac{1}{6}$. Then,

$$E(x) = \sum_x x f(x) = 1.\frac{1}{6} + \dots + 6.\frac{1}{6} = 3.5$$

Suppose X has the **geometric distribution** with the parameter p i.e. x is the number of Bern(p) trials until you obtain your first success (e.g. $FFFS$ gives $x = 4$).

$$f(x) = (1 - p)^{x-1}p, \quad x = 1, 2, \dots$$

Notation: $X \sim \text{Geom}(p)$

Suppose I take independent foul shots but the chance of making any particular shot is only 0.4. What's the probability that it will take me at least 3 tries to make successful shot?

The number of tries until my first success is $X \sim \text{Geom}(0.4)$. Thus,

$$\begin{aligned} P(X \geq 3) &= 1 - P(X \leq 2) \\ &= 1 - P(X = 1) - P(X = 2) \\ &= 1 - 0.4 - 0.6 * 0.4 \\ &= 0.36 \end{aligned}$$

Now, let's find the **expected value** of $X \sim \text{Geom}(p)$

$$\begin{aligned}
E(X) &= \sum_x x f(x) \\
&= \sum_{x=1}^{\infty} x q^{x-1} p \quad (\text{where } q = 1 - p) \\
&= p \sum_{x=1}^{\infty} \frac{d}{dq} q^x \\
&= p \frac{d}{dq} \sum_{x=1}^{\infty} q^x \quad (\text{swap derivative and sum}) \\
&= p \frac{d}{dq} \frac{q}{1-q} \quad (\text{geometric sum}) \\
&= p \left\{ \frac{(1-q) - q(-1)}{(1-q)^2} \right\} \\
&= \frac{1}{p}
\end{aligned}$$

Example: $X \sim \text{Exp}(\lambda)$. $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$. Then,

$$\begin{aligned}
E(X) &= \int_{\mathbb{R}} x f(x) dx \\
&= \int_0^{\infty} x \lambda e^{-\lambda x} dx \\
&= -x e^{-\lambda x} \Big|_0^{\infty} - \int_0^{\infty} (-e^{-\lambda x}) dx \quad (\text{by parts}) \\
&= \int_0^{\infty} e^{-\lambda x} dx \quad (\text{L' H\^opital rule}) \\
&= \frac{1}{\lambda}
\end{aligned}$$

3.1.4 Law of the unconscious statistician (LOTUS)

Theorem: The expected value of a function of X, say $h(x)$ is,

$$E[h(X)] = \begin{cases} \sum_x h(x) f(x) & \text{if } X \text{ is discrete,} \\ \int_{\mathbb{R}} h(x) f(x) dx & \text{if } X \text{ is continuous.} \end{cases}$$

$E[h(x)]$ is weighted function of $h(x)$ where the weights are $f(x)$'s.

Remark: It looks like a definition, but it's really a theorem - that's why they call it LOTUS.

Example: $E[\sin x] = \int_{\mathbb{R}} \sin x f(x) dx$

Definition: The k^{th} moment is

$$E[X^k] = \begin{cases} \sum_x x^k f(x) & \text{if } X \text{ is discrete} \\ \int_{\mathbb{R}} x^k f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Example: Suppose $X \sim \text{Bern}(p)$ so that $f(1) = p$ and $f(0) = q$.

$$E[X^k] = \sum_x x^k f(x) = 0^k q + 1^k p = p \quad \text{for all } k!$$

Example: Suppose $X \sim \text{Exp}(\lambda)$. $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$ then,

$$\begin{aligned} E[X^k] &= \int_{\mathbb{R}} x^k f(x) dx \\ &= \int_0^{\infty} x^k \lambda e^{-\lambda x} dx \\ &= \int_0^{\infty} \left(\frac{y}{\lambda}\right)^k \lambda e^{-\lambda \frac{y}{\lambda}} \frac{1}{\lambda} dy \quad (\text{substitute } y = \lambda x) \\ &= \frac{1}{\lambda^k} \int_0^{\infty} y^{k+1-1} e^{-y} dy \\ &= \frac{\Gamma(k+1)}{\lambda^k} \quad (\text{by definition of gamma function}) \\ &= \frac{k!}{\lambda^k} \end{aligned}$$

Definition: The k^{th} **central moment** of X is

$$E[(X - \mu)^k] = \begin{cases} \sum_x (x - \mu)^k f(x) & X \text{ is discrete} \\ \int_{\mathbb{R}} (x - \mu)^k f(x) dx & X \text{ is continuous} \end{cases}$$

Definition: The **variance** of X is the second central moment i.e. the expected squared deviation of X from its mean.

$$\text{Var}(X) = E[(X - \mu)^2]$$

Notation: $\sigma^2 = \text{Var}(X)$

Definition: The **standard deviation** of X is $\sigma = +\sqrt{\text{var}(x)}$

Example: $X \sim \text{Bern}(p)$ so that $f(1) = p$, $f(0) = q = 1 - p$

$$\mu = E[X] = p \quad \text{then,}$$

$$\begin{aligned}
\text{Var}[X] &= E[(X - \mu)^2] \\
&= \sum_x (x - p)^2 P(X = x) \\
&= (0 - p)^2 \cdot q + (1 - p)^2 \cdot p \\
&= p^2 q + q^2 p \\
&= pq(p + q) \\
&= pq \quad (\text{since } p + q = 1)
\end{aligned}$$

Theorem: For any $h(x)$ and constants a and b - “shift happens”,

$$E[ah(X) + b] = aE[h(X)] + b$$

Proof:

$$\begin{aligned}
E[ah(X) + b] &= \int_{\mathbb{R}} (ah(x) + b)f(x)dx \\
&= a \int_{\mathbb{R}} h(x)f(x)dx + b \int_{\mathbb{R}} f(x)dx \\
&= aE[h(x)] + b
\end{aligned}$$

Corollary: In particular,

$$E[aX + b] = aE[X] + b$$

Theorem (Easier way to calculate variance):

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Proof:

$$\begin{aligned}
\text{Var}(X) &= E[(X - \mu)^2] \\
&= E[X^2 - 2\mu X + \mu^2] \\
&= E[X^2] - 2\mu E[X] + \mu^2 \\
&= E[X^2] - \mu^2 \quad \text{where } E[X] = \mu
\end{aligned}$$

Example: Suppose $X \sim \text{Bern}(p)$. Recall that $E[X^k] = p$ for all $k = 1, 2, \dots$. Then,

$$\begin{aligned}
\text{Var}[X] &= E[X^2] - (E[X])^2 \\
&= p - p^2 \\
&= p \cdot q
\end{aligned}$$

Example: $X \sim \text{Unif}(a, b)$. $f(x) = \frac{1}{b-a}$, $a < x < b$ then,

$$E[X] = \int_{\mathbb{R}} f(x)dx = \int_a^b \frac{x}{b-a}dx = \frac{a+b}{2}$$

$$E[X^2] = \int_{\mathbb{R}} xf(x)dx = \int_a^b \frac{x^2}{b-a}dx = \frac{a^2 + ab + b^2}{3}$$

$$Var(X) = E[X^2] - (E[X])^2 = \frac{(a-b)^2}{12}$$

Theorem: Variance doesn't put up with shift b.

$$Var(aX + b) = a^2 \cdot Var(X)$$

Proof:

$$\begin{aligned} Var(aX + b) &= E[(aX + b)^2] - (E[aX + b])^2 \\ &= E[a^2X^2 + b^2 + 2abX] - (aE[X] + b)^2 \\ &= a^2E[X^2] + b^2 + 2abE[X] - a^2(E[X])^2 - 2abE[X] - b^2 \\ &= a^2\{E[X^2] - (E[X])^2\} \\ &= a^2Var(X) \end{aligned}$$

Example: $X \sim Bern(0.3)$

$$E[X] = p = 0.3$$

$$Var[X] = pq = 0.3 * 0.7 = 0.21$$

Let

$$Y = h(x) = 4x + 5 \text{ then,}$$

$$E[Y] = E[4X + 5] = 4E[X] + 5 = 6.2$$

$$Var[Y] = Var[4X + 5] = 16Var[X] = 3.36$$

Approximations to $E[h(x)]$ and $Var[h(x)]$

Sometimes $Y = h(x)$ is messy and we may have to approximate $E[h(x)]$ and $Var[h(x)]$ via a Taylor series approach. Let $\mu = E[X]$ and $\sigma^2 = Var(X)$ and note that

$$Y = h(\mu) + (X - \mu) \cdot h'(\mu) + \frac{(X - \mu)^2}{2} \cdot h''(\mu) + R$$

where, R is remainder term that we will ignore. Then,

$$E[Y] = h(\mu) + E[X - \mu] \cdot h'(\mu) + \frac{E[(X - \mu)^2]}{2} \cdot h''(\mu) = h(\mu) + \frac{h''(\mu)\sigma^2}{2}$$

and (now an even-crude approximation)

$$\text{Var}(Y) = \text{Var}[h(\mu) + (X - \mu) \cdot h'(\mu)] = [h'(\mu)]^2 \sigma^2$$

Example: Suppose X has pdf $f(x) = 3x^2$, $0 \leq x \leq 1$ and we want to test out our approximations on the “complicated” random variable $Y = h(X) = X^{\frac{3}{4}}$.

$$E[Y] = \int_{\mathbb{R}} x^{\frac{3}{4}} \cdot f(x) dx = \int_0^1 3x^{\frac{11}{4}} dx = \frac{4}{5}$$

$$E[Y^2] = \int_{\mathbb{R}} x^{\frac{6}{4}} f(x) dx = \int_0^1 3x^{\frac{7}{2}} dx = \frac{2}{3}$$

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = 0.0267$$

Before approximation, note that,

$$\mu = E[X] = \int_{\mathbb{R}} x f(x) dx = \int_0^1 3x^3 dx = \frac{3}{4}$$

$$E[X^2] = \int_{\mathbb{R}} x^2 f(x) dx = \int_0^1 3x^4 dx = \frac{3}{5}$$

$$\sigma^2 = \text{Var}[X] = E[X^2] - (E[X])^2 = 0.0375$$

$$h(\mu) = \mu^{\frac{3}{4}} = \left(\frac{3}{4}\right)^{\frac{3}{4}} = 0.8059$$

$$h'(\mu) = \frac{3}{4} \cdot \mu^{-\frac{1}{4}} = 0.8059$$

$$h''(\mu) = -\left(\frac{3}{16}\right) \cdot \mu^{-\frac{5}{4}} = -0.2686$$

$$E[Y] = h(\mu) + \frac{h''(\mu)\sigma^2}{2} = 0.8009$$

$$\text{Var}(Y) = [h'(\mu)]^2 \cdot \sigma^2 = 0.0243$$

Moment Generating Functions

Definition: The **moment generating function** (mgf) of RV of X is

$$M_X(t) = E[e^{tX}]$$

Remark: $M_X(t)$ is a function of t and not of X.

Example: $X \sim \text{Bern}(p)$ so that $X = 1$ with probability p and 0 with probability q .

$$\begin{aligned} M_X(t) &= E[e^{tX}] \\ &= \sum_x e^{tx} f(x) \\ &= e^{t \cdot 1} \cdot p + e^{t \cdot 0} q \\ M_X(t) &= p \cdot e^t + q \end{aligned}$$

Example: If $X \sim \text{Exp}(\lambda)$, then,

$$\begin{aligned} M_x(t) &= E[e^{tX}] \\ &= \int_{\mathbb{R}} e^{tX} f(x) dx \quad (\text{LOTUS}) \\ &= \int_0^{\infty} e^{tX} \lambda e^{-\lambda x} dx \\ &= \lambda \int_0^{\infty} e^{(t-\lambda)x} dx \\ &= \frac{\lambda}{\lambda - t} \quad \lambda > t \end{aligned}$$

Big theorem: Under certain conditions (e.g. $M_X(t)$) must exist for all $t \in (-\epsilon, \epsilon)$ for some $\epsilon > 0$, we have

$$E[X^k] = \left. \frac{d^k}{dt^k} M_X(t) \right|_{t=0}, \quad k = 1, 2, \dots$$

Thus, we can generate the moments of X from mgf. Sometimes it's easier to get moments this way directly.

Proof:

$$\begin{aligned} M_X(t) &= E[e^{tX}] \\ &= E\left[\sum_{k=0}^{\infty} \frac{(tX)^k}{k!}\right] \quad (\text{McLaurin Series}) \\ &= \sum_{k=0}^{\infty} E\left[\frac{(tX)^k}{k!}\right] \\ &= 1 + tE(X) + \frac{t^2 E[X^2]}{2} + \dots \end{aligned}$$

This implies,

$$\frac{d}{dt} M_X(t) = E[X] + tE[X^2] + \dots$$

and so,

$$\left. \frac{d}{dt} M_X(t) \right|_{t=0} = E[X]$$

Example: $X \sim \text{Bern}(p)$. Then $M_x(t) = pe^t + q$ and

$$\begin{aligned} E[X] &= \left. \frac{d}{dt} M_x(t) \right|_{t=0} \\ &= \left. \frac{d}{dt} (pe^t + q) \right|_{t=0} \\ &= pe^t \Big|_{t=0} \\ &= p \end{aligned}$$

In fact, it's easy to see that $E[X^k] = \left. \frac{d^k}{dt^k} M_x(t) \right|_{t=0} = p$ for all k .

Example: $X \sim \text{Exp}(\lambda)$. Then $M_x(t) = \frac{\lambda}{\lambda - t}$ for $\lambda > t$. So,

$$E[X] = \left. \frac{d}{dt} M_x(t) \right|_{t=0} = \left. \frac{\lambda}{(\lambda - t)^2} \right|_{t=0} = \frac{1}{\lambda}$$

$$E[X^2] = \left. \frac{d^2}{dt^2} M_x(t) \right|_{t=0} = \left. \frac{2\lambda}{(\lambda - t)^3} \right|_{t=0} = \frac{2}{\lambda^2}$$

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

Theorem (mgf of linear function of X): Suppose X has mgf $M_x(t)$ and let $Y = aX + b$.

$$M_Y(t) = e^{tb} M_X(at)$$

Proof:

$$\begin{aligned} M_Y(t) &= E[e^{tY}] \\ &= E[e^{t(ax+b)}] \\ &= e^{tb} E[e^{t(ax)}] \\ &= e^{tb} M_X(at) \end{aligned}$$

Example: Let $X \sim \text{Exp}(\lambda)$ and $Y = 3X + 2$. Then,

$$\begin{aligned} M_Y(t) &= e^{2t} M_X(3t) \\ &= e^{2t} \frac{\lambda}{\lambda - 3t} \quad \text{if } \lambda > 3t \end{aligned}$$

Theorem (identifying distribution): In this class, each distribution has a unique mgf.

Proof: Not here!

Example: Suppose that Y has mgf,

$$\begin{aligned} M_Y(t) &= e^{2t} M_X(3t) \\ &= e^{2t} \frac{\lambda}{\lambda - 3t} \quad \text{if } \lambda > 3t \end{aligned}$$

Then by previous example and uniqueness of Mgf's, it must be the case that $Y \sim 3X + 2$, where $X \sim \text{Exp}(\lambda)$.

3.1.5 Some Probability Inequalities

Theorem: Markov's Inequality

If X is non-negative random variable and $c > 0$ then $P(X \geq c) \leq E[X]/C$.

Proof: Because X is non-negative, we have,

$$\begin{aligned} E[X] &= \int_{\mathbb{R}} x f(x) dx \\ &= \int_0^{\infty} x f(x) dx \\ &\geq \int_c^{\infty} x f(x) dx \\ &\geq c \int_c^{\infty} f(x) dx \\ &= c \cdot P(X \geq c) \end{aligned}$$

Theorem: Chebychev's Inequality

Suppose $E[X] = \mu$ and $\text{Var}[X] = \sigma^2$. Then, for any $c > 0$, $P(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$

Proof: By Markov with $|X - \mu|^2$ in place of X and c^2 in place of c , we have,

$$\begin{aligned} P(|X - \mu| \geq c) &= P((X - \mu)^2 \geq c^2) \\ &\leq \frac{E[(X - \mu)^2]}{c^2} \\ &\quad \{ |X - \mu| \geq c \text{ if and only if } (X - \mu)^2 \geq c^2 \} \\ &= \frac{\sigma^2}{c^2} \end{aligned}$$

Remark: Can also write $P(|X - \mu| < c) \geq 1 - \frac{\sigma^2}{c^2}$. If $c = k \cdot \sigma$ then $P(|X - \mu| \geq k \cdot \sigma) \leq \frac{1}{k^2}$.

It means if you move further out in terms of standard deviation, the smaller the probability will be. Chebychev gives a bound on the probability that X deviates from the mean by more than a constant in terms of constant and the variance. You can always use Chebychev but it's crude.

Example: $X \sim Unif(0, 1)$ $f(x) = 1$ for $0 < x < 1$.

Recall that $E[X] = \frac{1}{2}$, $Var(X) = \frac{1}{12}$.

Chebychev implies that

$$P(|X - \frac{1}{2}| \geq c) \leq \frac{1}{12c^2}$$

In particular, for $c = \frac{1}{3}$,

$$P(|X - \frac{1}{2}| \geq \frac{1}{3}) \leq \frac{3}{4} \text{ (upper bound)}$$

Let's compare the upper bound to exact answer,

$$\begin{aligned} P(|x - \frac{1}{2}| \geq \frac{1}{3}) &= 1 - P(|x - \frac{1}{2}| < \frac{1}{3}) \\ &= 1 - P(-\frac{1}{3} < x - \frac{1}{2} < \frac{1}{3}) \\ &= 1 - P(\frac{1}{6} < x < \frac{5}{6}) \\ &= 1 - \int_{1/6}^{5/6} f(x) dx \\ &= 1 - \frac{2}{3} \\ &= \frac{1}{3} \end{aligned}$$

So, Chebychev bound of $\frac{3}{4}$ is pretty high in comparison.

Theorem (Chernoff's inequality): For any c ,

$$P(X \geq C) \leq e^{-ct} M_X(t)$$

Proof: By Markov with e^{tx} in place of X and e^{tc} in place of c , we have,

$$\begin{aligned} P(X \geq C) &= P(e^{tx} \geq e^{tc}) \\ &= e^{-tc} E[e^{tX}] \text{ from Markov's inequality, } P(Y \geq a) \leq \frac{E[Y]}{a} \\ &= e^{-ct} M_X(t) \end{aligned}$$

Example: Suppose X has the standard normal distribution with pdf $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ for $x \in \mathbb{R}$. It is easy to show that mgf of standard normal is

$$\begin{aligned} M_x(t) &= E[e^{tX}] \\ &= \int_{\mathbb{R}} e^{tx} \phi(x) dx \\ &= e^{\frac{t^2}{2}} \end{aligned}$$

Then using Chernoff with $t = c$ immediately yields the tail probability.

$$\begin{aligned} P(X \geq C) &\leq e^{-C^2} M_X(c) \\ &= e^{-\frac{c^2}{2}} \end{aligned}$$

3.1.6 Functions of Random Variable

Problem: You have RV X and you know its pmf/pdf $f(x)$.

Define $Y = h(x)$ (Some function of X). Find $g(y)$ the pmf/pdf of X .

Remark: Recall that LOTUS gave us results for $E[h(x)]$. But this is much more general than LOTUS, because we are going to get entire distribution of $h(X)$.

Discrete Case: X discrete implies Y discrete.

$$\begin{aligned} g(y) &= P(Y = y) \\ &= P(h(X) = y) \\ &= P(x|h(x) = y) \quad (\text{Probability of } x\text{'s such that } h(x) = y) \\ &= \sum_{x|h(x)=y} f(x) \end{aligned}$$

Example: X is the number of H's in 2 coin tosses. We want the pmf of $Y = h(x) = x^3 - x$.

{TT, TH, HT, HH}

	x	0	1	2
$f(x) = P(X = x)$	$f(x) = P(X = x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
	$y = x^3 - x$	0	0	6

3 values of x map into 2 values of y i.e. 0 & 6.

$$\begin{aligned} g(0) &= P(Y = 0) = P(X = 0 \text{ or } 1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} \\ g(6) &= P(Y = 6) = P(X = 2) = \frac{1}{4} \end{aligned}$$

$$g(y) = \begin{cases} \frac{3}{4} & \text{if } y = 0 \\ \frac{1}{4} & \text{if } y = 6 \end{cases}$$

Example: X is discrete with

$$f(x) = \begin{cases} \frac{1}{8} & \text{if } x = -1 \\ \frac{3}{8} & \text{if } x = 0 \\ \frac{1}{3} & \text{if } x = 1 \\ \frac{1}{6} & \text{if } x = 2 \end{cases}$$

Let $Y = X^2$ (so Y can equal to 0, 1 or 4).

$$g(y) = \begin{cases} P(Y = 0) = f(0) = \frac{3}{8} \\ P(Y = 1) = f(1) + f(-1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4} \\ P(Y = 4) = f(2) = \frac{1}{6} \end{cases}$$

Continuous Case: X is continuous implies Y can be continuous/discrete.

Example: $Y = X^2$ (clearly continuous)

Example:

$$Y = \begin{cases} 0 & \text{if } X < 0 \\ 1 & \text{if } X \geq 0 \end{cases} \quad \text{is not continuous}$$

Method: Compute $G(y)$, the cdf of Y ,

$$\begin{aligned} G(y) &= P(Y \leq y) = P(h(x) \leq y) \\ &= \int_{\{x|h(x) \leq y\}} f(x) dx \end{aligned}$$

If $G(y)$ is continuous, construct the pdf $g(y)$ by differentiating.

Example: $f(x) = |x|, -1 \leq x \leq 1$. Find the pdf of RV $Y = h(X) = x^2$.

$$\begin{aligned} G(y) &= P(Y \leq y) \\ &= P(X^2 \leq y) = \begin{cases} 0 & \text{if } y \leq 0 \\ 1 & \text{if } y \geq 1 \\ (*) & \text{if } 0 < y < 1 \end{cases} \quad x^2 \text{ must be between 0 and 1} \end{aligned}$$

where,

$$\begin{aligned} * &= P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} |x| dx \\ &= y \end{aligned}$$

Thus,

$$G(y) = \begin{cases} 0 & \text{if } y \leq 0 \\ 1 & \text{if } y \geq 1 \\ y & \text{if } 0 < y < 1 \end{cases}$$

This implies,

$$g(y) = G'(y) = \begin{cases} 0 & \text{if } y < 0 \text{ and } y \geq 1 \\ 1 & \text{if } 0 < y < 1 \end{cases}$$

This means Y has the $Unif(0, 1)$ distribution.

Explanation

Since, $Y = X^2$ so, $0 \leq Y \leq 1$. The cdf is determined as

* If $y \leq 0$ then $P(X^2 \leq y) = 0$ because X^2 is non-negative.

* If $y \geq 1$ then $P(X^2 \leq y) = 1$ because X^2 is at most 1.

* If $0 < y < 1$, the $P(-\sqrt{y} \leq X \leq \sqrt{y})$

Example: Suppose $U \sim Unif(0, 1)$ Find the pdf of $Y = -\ln(1 - U)$.

$$\begin{aligned} G(y) &= P(Y \leq y) \\ &= P(-\ln(1 - U) \leq y) \\ &= P(1 - U \leq e^{-y}) \\ &= \int_0^{1-e^{-y}} f(u) du \\ &= 1 - e^{-y} \quad (\text{since } f(u) = 1) \end{aligned}$$

$$g(y) = G'(y) = e^{-y}, y > 0$$

This implies $Y \sim Exp(\lambda = 1)$.

$$G(y) = \begin{cases} 0 & \text{if } y < 0 \\ 1 - e^{-y} & \text{if } y \geq 0 \end{cases}$$

For $y < 0$, $P(Y \leq y) = 0$ because $Y = -\ln(1 - U)$ is non-negative.

3.1.7 Inverse Transform Theorem/Probability Integral Transform

Suppose X is a continuous random variable having cdf $F(x)$. Then the random variable $F(X) \sim Unif(0, 1)$.

Proof: Let $Y = F(X)$. The cdf of Y is

$$\begin{aligned}
 G(y) &= P(Y \leq y) \\
 &= P(F(X) \leq y) \\
 &= P(X \leq F^{-1}(y)) \quad \{\text{cdf is monotonically increasing}\} \\
 &= F(F^{-1}(y)) \\
 &= y
 \end{aligned}$$

Monotonically increasing means as input increases, output never decreases. Example: $f(x) = x^2$, $x \geq 0$.

Remark: This is a great theorem since it applies to all continuous RVs X .

Corollary: $X = F^{-1}(U)$ so that you can plug $Unif(0, 1)$ RV into the inverse cdf to generate a realization of RV having X 's distribution.

Method: Set $F(X) = U$ and solve $X = F^{-1}(U)$ to generate X .

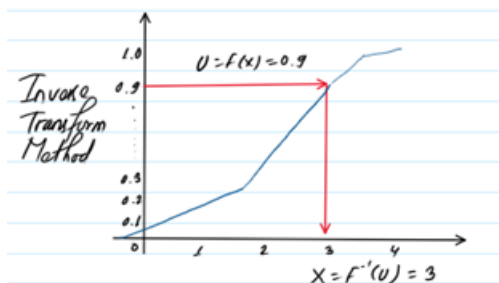


Figure 2: Inverse function

Example: Suppose X is $Exp(\lambda)$ so that it has cdf $F(x) = 1 - e^{-\lambda x}$. Similar to previous example, set $F(x) = 1 - e^{-\lambda x} = U$ and generate an $Exp(\lambda)$ RV by solving for,

Remark: If you'd like to generate a nice, beautiful $Exp(\lambda)$ pdf on a computer then

- Generate 10000 $Unif(0, 1)$. Use rand function in excel or unifrnd in Matlab).
- Plug those 10000 into equation for X above and
- Plot the histogram of X .

Another way to find pdf of a function of a continuous RV

Suppose that $Y = h(x)$ is a monotonic function of a continuous RV X having pdf $f(x)$ and cdf $F(x)$. Let's get the pdf $g(y)$ of Y directly.

$$\begin{aligned}
g(y) &= \frac{d}{dy} G(y) \\
&= \frac{d}{dy} P(Y \leq y) \\
&= \frac{d}{dy} P(X \leq h^{-1}(y)) \quad (h(x) \text{ is monotonic}) \\
&= \frac{d}{dy} F(h^{-1}(y)) \\
&= f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right| \quad (\text{chain rule})
\end{aligned}$$

Example: Suppose that $f(x) = 3x^2$, $0 < x < 1$. Let $Y = h(x) = x^{1/2}$ which is monotone increasing.

$$\begin{aligned}
g(y) &= f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right| \\
&= f(y^2) \left| \frac{d(y^2)}{dy} \right| \\
&= 3y^4 \cdot 2y \\
&= 6y^5, \quad 0 < y < 1
\end{aligned}$$

Explanation

$$Y = X^{1/2}$$

$$X = Y^2$$

$h^{-1}(y) = X = Y^2$ $h^{-1}(y)$ is inverse function of $h(x)$ expressed in terms of y

$$f(h^{-1}(y)) = f(y^2) = 3y^4$$

Theorem (why LOTUS works): Let us assume $h(\cdot)$ is monotonically increasing. Then

$$\begin{aligned}
E[h(x)] &= E[Y] \\
&= \int_R y f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right| dy \\
&= \int_R h(x) f(x) \left| \frac{dx}{dy} \right| dy \\
&= \int_R h(x) f(x) dx
\end{aligned}$$

4. Bivariate Random Variable

We will look at two random variables simultaneously.

Example: Choose a person at random. Look at their height and weight (X, Y) . Obviously, X and Y will be related somehow.

4.1 Joint Distribution

4.1.1 Discrete Random Variable

Definition: If X and Y are discrete random variables then (X, Y) is called **jointly discrete bivariate random variable**.

The joint (or bivariate) pmf is

$$f(x, y) = P(X = x, Y = y), \quad \forall x, y$$

Properties

- $0 \leq f(x, y) \leq 1$
- $\sum_x \sum_y f(x, y) = 1$
- $A \subseteq \mathbb{R} \implies P((X, Y) \in A) = \sum \sum_{(x, y) \in A} f(x, y)$

Example: 3 Socks in a box numbered 1,2,3. Draw 2 socks at random without replacement. X = number of the first sock, Y = number of the second sock. The joint pmf $f(x, y)$ is

$f(x, y)$	$x = 1$	$x = 2$	$x = 3$	$P(Y = y)$
$y = 1$	0	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{3}$
$y = 2$	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{1}{3}$
$y = 3$	$\frac{1}{6}$	$\frac{1}{6}$	0	$\frac{1}{3}$
$P(X = x)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1

- Diagonal entries $X=Y$ is 0 because you can't pick same sock twice.
- Each non-diagonal entry has probability $\frac{1}{6}$ since there are $3! = 6$ ways to select socks.

$f_x(x) = P(X = x)$ is the **marginal pmf of X**. $f_y(y) = P(Y = y)$ is the **marginal pmf of Y**.

By the law of Total Probability:

$$P(X = 1) = \sum_{y=1}^3 P(X = 1, Y = y) = \frac{1}{3}$$

In addition;

$$\begin{aligned}
 P(X > 2, Y \geq 2) &= \sum_{x \geq 2} \sum_{y \geq 2} f(x, y) \\
 &= f(2, 2) + f(2, 3) + f(3, 2) + f(3, 3) \\
 &= 0 + \frac{1}{6} + \frac{1}{6} + 0 \\
 &= \frac{1}{3}
 \end{aligned}$$

4.1.2 Continuous Random Variable

Definition If X and Y are continuous random variables then (X, Y) is a **jointly continuous bivariate random variable** if there exists a magic function $f(x, y)$ such that

- $f(x, y) \geq 0, \forall x, y$
- $\int \int_{\mathbb{R}^2} f(x, y) dx dy = 1$
- $P(A) = P((X, Y) \in A) = \int \int_A f(x, y) dx dy$

In this case, $f(x, y)$ is called the **joint pdf**. If $A \subseteq \mathbb{R}^2$ then $P(A)$ is the volume between $f(x, y)$ and A . Think of $f(x, y) dx dy \approx P(x < X < x + dx, y < Y < y + dy)$

Example: Choose a point (X, Y) at random in the interior of the circle inscribed in the unit square $C = (x - \frac{1}{2})^2 + (y - \frac{1}{2})^2 \leq \frac{1}{4}$. Find the pdf (X, Y) .

Area of circle is

$$\pi \left(\frac{1}{2}\right)^2 = \frac{\pi}{4}$$

$$f(x, y) = \begin{cases} \frac{4}{\pi} & \text{if } (x, y) \in C \\ 0 & \text{if otherwise} \end{cases}$$

$f(x, y) = \frac{4}{\pi}$ because if you integrate $\frac{4}{\pi}$ in the region of circle you get 1.

Application: Toss n darts randomly into the unit square. The probability that any individual dart will land in the circle is $\frac{\pi}{4}$. It stands to reason that the proportion of darts $\hat{P}_n$ that land in the circle will be approximately $\frac{\pi}{4}$. So you can use $4\hat{P}_n$ to estimate π .

Example: Suppose that

$$f(x, y) = \begin{cases} 4xy & \text{if } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the probability (volume) of the region $0 \leq y \leq 1 - x^2$.

$$\begin{aligned} V &= \int_0^1 \int_0^{1-x^2} 4xy dy dx \\ &= \int_0^1 \int_0^{\sqrt{1-y}} 4xy dx dy \\ &= \frac{1}{3} \end{aligned}$$

4.2 Cumulative Distribution Function

Definition: The **joint (bivariate cdf)** of X and Y is

$$F(X, Y) = P(X \leq x, Y \leq y) \text{ for all } x, y$$

$$F(X, Y) = \begin{cases} \sum \sum_{s \leq x, t \leq y} f(s, t) & \text{discrete} \\ \int_{-\infty}^y \int_{-\infty}^x f(s, t) ds dt & \text{continuous} \end{cases}$$

Going from CDF's to PDF's (continuous case): 1-dimension: $f(x) = F'(x) = \frac{d}{dx} \int_{-\infty}^x f(t) dt$

2-dimension; $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y) = \frac{\partial^2}{\partial x \partial y} \int_{-\infty}^x \int_{-\infty}^y f(s, t) dt ds$

Properties:

- $F(X, Y)$ is non-decreasing in both x and y .
- $\lim_{x \rightarrow -\infty} F(x, y) = \lim_{y \rightarrow -\infty} F(x, y) = 0$
- $\lim_{x \rightarrow -\infty} F(x, y) = F_Y(y) = P(Y \leq y)$ marginal CDF of Y
- $\lim_{y \rightarrow -\infty} F(x, y) = F_X(x) = P(X \leq x)$ marginal CDF of X
- $\lim_{x \rightarrow -\infty} \lim_{y \rightarrow -\infty} F(x, y) = 1$
- $F(x, y)$ is continuous from the right in both x and y .

Example: Suppose

$$F(x, y) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-(x+y)} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{if } x < 0 \text{ or } y < 0 \end{cases}$$

The marginal CDF of X is

$$F_X(x) = \lim_{y \rightarrow \infty} F(x, y) = \begin{cases} 1 - e^{-x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

The Joint PDF is

$$\begin{aligned} f(x, y) &= \frac{\partial^2}{\partial x \partial y} F(x, y) \\ &= \frac{\partial}{\partial y} (e^{-x} - e^{-y} e^{-x}) \\ &= e^{-(x+y)} \quad \text{if } x \geq 0, y \geq 0 \end{aligned}$$

Marginal distribution from Joint,

$$f_X(x) = \int_0^\infty f(x, y) dy$$

4.3 Marginal Distribution

Definition: If X and Y are jointly discrete, then **marginal PMF's** of X and Y are respectively,

$$f_X(x) = P(X = x) = \sum_y f(x, y)$$

&

$$f_Y(y) = P(Y = y) = \sum_x f(x, y)$$

Example (Discrete Case): $f(x, y) = P(X = x, Y = y)$

$f(x, y)$	$X = 1$	$X = 2$	$X = 3$	$P(Y = y)$
$Y = 40$	0.1	0.07	0.12	0.2
$Y = 60$	0.29	0.03	0.48	0.8
$P(X = x)$	0.3	0.1	0.6	1

By Total Probability,

$$P(X = 1) = P(X = 1, Y = \text{any}) = 0.3$$

Definition: If X and Y are **jointly continuous**, then the **marginal PDF's** of X and Y are

$$\begin{aligned} f_X(x) &= \int_{\mathbb{R}} f(x, y) dy \\ f_Y(y) &= \int_{\mathbb{R}} f(x, y) dx \end{aligned}$$

Example:

$$f(x, y) = \begin{cases} e^{-(x+y)} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Then marginal PDF of X is

$$f_X(x) = \int_{\mathbb{R}} f(x, y) dy = \int_0^{\infty} e^{-(x+y)} dy = e^{-x} \text{ if } x \geq 0$$

Example:

$$f(x, y) = \begin{cases} \frac{21}{4}x^2y & \text{if } x^2 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Note **funny limits** where PDF is positive i.e., $x^2 \leq y \leq 1$.

$$\begin{aligned} f_X(x) &= \int_{\mathbb{R}} f(x, y) dy \\ &= \int_{x^2}^1 \frac{21}{4}x^2y dy \\ &= \frac{21}{8}x^2(1 - x^4), -1 \leq x \leq 1 \end{aligned}$$

If we integrate $\frac{21}{8}x^2(1 - x^4)$ from -1 to 1 , it must integrate to 1 because we know legitimate PDF $f(x)$ must integrate to 1.

4.4 Conditional Distributions

Recall the conditional probability,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ if } P(B) > 0$$

Suppose that X and Y are jointly discrete RVs. Then if $P(X = x) > 0$.

$$P(Y = y|X = x) = \frac{P(X = x \cap Y = y)}{P(X = x)} = \frac{f(x, y)}{f_X(x)}$$

Definition: If $f_X(x) > 0$ then the **conditional PMF/PDF of Y given $X=x$** is

$$f_{Y/X}(y/x) = \frac{f(x, y)}{f_X(x)}$$

Remark: We usually just write $f(y|x)$ instead of $f_{y|x}(Y|x)$.

$f(x, y)$	$X = 1$	$X = 2$	$X = 3$	$f_Y(y)$
$Y = 40$	0.1	0.07	0.12	0.2
$Y = 60$	0.29	0.03	0.48	0.8
$f_X(x)$	0.3	0.1	0.6	1

Discrete Example:

$$f(x|y=60) = \frac{f(x, 60)}{f_Y(60)} = \frac{f(x, 60)}{0.8} = \begin{cases} \frac{29}{80} & \text{if } x = 1 \\ \frac{3}{80} & \text{if } x = 2 \\ \frac{48}{80} & \text{if } x = 3 \end{cases}$$

Continuous Example:

$$\begin{aligned} f(x, y) &= \frac{21}{4}x^2y \quad \text{if } x^2 \leq y \leq 1 \\ f_X(x) &= \frac{21}{8}x^2(1 - x^4) \quad \text{if } -1 \leq x \leq 1 \\ f_Y(y) &= \frac{7}{2}y^{\frac{5}{2}} \quad \text{if } 0 \leq y \leq 1 \end{aligned}$$

Then the conditional PDF of Y given $X = x$ is

$$\begin{aligned} f(y|x) &= \frac{f(x, y)}{f_X(x)} \\ &= \frac{\frac{21}{4}x^2y}{\frac{21}{8}x^2(1 - x^4)} \\ &= \frac{2y}{1 - x^4} \quad \text{if } x^2 \leq y \leq 1 \end{aligned}$$

If you integrate $f(y|x)$ over x^2 to 1 you get 1.

$$\begin{aligned} f(y|\frac{1}{2}) &= \frac{f(\frac{1}{2}, y)}{f_X(\frac{1}{2})} \\ &= \frac{\frac{21}{4} \cdot \frac{1}{4} \cdot y}{\frac{21}{8} \cdot \frac{1}{4} \cdot (1 - \frac{1}{16})} \\ &= \frac{32}{15}y \quad \text{if } \frac{1}{4} \leq y \leq 1 \end{aligned}$$

Note that $\frac{2}{1-x^4}$ is a constant with respect to y , we can check to see that $f(y|x)$ is a legit conditional PDF.

$$\int_{\mathbb{R}} f(y|x)dy = \int_{x^2}^1 \frac{2y}{1 - x^4}dy = 1$$

Typical Problem: Given $f_X(X)$, $f_Y(y)$ and $f(Y|X)$.

Game Plan: Find $f(x, y) = f_X(x)f(y|x)$ and then $f_Y(y) = \int_{\mathbb{R}} f(x, y)dx$

Example: Suppose $f_X(x) = 2x$ for $0 < x < 1$. Given $X = x$ suppose that $Y|x \sim Unif[0, x]$. Now find $f_Y(y)$.

$Y|x \sim Unif[0, x]$ implies that $f(y|x) = \frac{1}{x}$ for $0 < y < x$. So,

$$\begin{aligned} f(x, y) &= f_X(x)f(y|x) \\ &= 2x \cdot \frac{1}{x} \quad \text{for } 0 < x < 1 \text{ and } 0 < y < x \\ &= 2 \quad 0 < y < x < 1 \end{aligned}$$

Thus,

$$\begin{aligned} f_Y(y) &= \int_{\mathbb{R}} f(x, y) dx \\ &= \int_y^1 2 dx \\ &= 2(1 - y) \quad 0 < y < 1 \end{aligned}$$

4.5 Independent Random Variable

Recall that two events are independent if $P(A \cap B) = P(A)P(B)$. Then,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

Similarly,

$$P(B|A) = P(B)$$

Definition: X and Y are **independent** random variables if for all x and y .

$$f(x, y) = f_X(x)f_Y(y)$$

4.5.1 Consequence of Independence

Defintion/Theorem (Two dimensional Unconscious Statistician): Let $h(x, y)$ be a function of RV's X and Y .

$$E[h(x, y)] = \begin{cases} \sum_x \sum_y h(x, y) f(x, y) & \text{discrete} \\ \int_{\mathbb{R}} \int_{\mathbb{R}} h(x, y) f(x, y) dx dy & \text{continuous} \end{cases}$$

Theorem: Whether or not X and Y are independent,

$$E[X + Y] = E[X] + E[Y]$$

Proof (Continuous case):

$$\begin{aligned}
E[X + Y] &= \int_{\mathbb{R}} \int_{\mathbb{R}} f(x, y) dx dy && \text{2-D LOTUS} \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} x f(x, y) dx dy + \int_{\mathbb{R}} \int_{\mathbb{R}} y f(x, y) dx dy \\
&= \int_{\mathbb{R}} x \int_{\mathbb{R}} f(x, y) dy + \int_{\mathbb{R}} y \int_{\mathbb{R}} f(x, y) dx && \text{Fubini's Theorem} \\
&= \int_{\mathbb{R}} x f_X(x) dx + \int_{\mathbb{R}} y f_Y(y) dy \\
&= E[X] + E[Y]
\end{aligned}$$

Corollary: If $X_1, X_2, \dots, X_n$ are RVs then

$$E[\sum_{i=1}^n X_i] = \sum_{i=1}^n E[X_i]$$

Theorem: If X and Y are independent, then $E[XY] = E[X]E[Y]$.

Proof (Continuous Case):

$$\begin{aligned}
E[XY] &= \int_{\mathbb{R}} \int_{\mathbb{R}} xy f(x, y) dx dy && \text{2-D LOTUS} \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} xy f_X(x) f_Y(y) dx dy && \text{X and Y are independent} \\
&= \left(\int_{\mathbb{R}} x f_X(x) dx \right) \left(\int_{\mathbb{R}} y f_Y(y) dy \right) \\
&= E[X]E[Y]
\end{aligned}$$

Remark: The above theorem is not necessarily true if X and Y are dependent.

Theorem: If X and Y are independent, then,

$$Var(X + Y) = var(X) + Var(Y)$$

Proof:

$$\begin{aligned}
Var(X + Y) &= E[(X + Y)^2] - (E[XY])^2 \\
&= E[X^2 + 2XY + Y^2] - (E[X] + E[Y])^2 \\
&= E[X^2] + 2E[XY] + E[Y^2] - \{(E[X])^2 + 2E[X]E[Y] + (E[Y])^2\} \\
&= E[X^2] + 2E[X]E[Y] + E[Y^2] - (E[X])^2 - 2E[X]E[Y] - (E[Y])^2 \\
&\quad \text{(since X and Y are independent)} \\
&= E[X^2] - (E[X])^2 + E[Y^2] - (E[Y])^2 \\
&= Var[X] + Var[Y]
\end{aligned}$$

Remark: The assumption of independence really is important here. If X and Y aren't independent, then the result might not hold.

Corollary: If $X_1, X_2, \dots, X_n$ are independent RVs then,

$$Var(\sum_{i=1}^n X_i) = \sum_{i=1}^n Var(X_i)$$

Proof: Induction

Corollary: If $X_1, X_2, \dots, X_n$ are independent RVs then,

$$Var(\sum_{i=1}^n a_i X_i + b) = \sum_{i=1}^n a_i^2 Var(X_i)$$

4.5.2 Random Samples

Definition: $X_1, X_2, \dots, X_n$ form a random sample if

- X_i 's are all independent.
- Each X_i has the same pmd/pdf $f(x)$.

Notation: $X_1, \dots, X_n \stackrel{iid}{\sim} f(x)$

Theorem: Suppose $X_1, \dots, X_n \stackrel{iid}{\sim} f(x)$ with $E[X_i] = \mu$ and $Var(X_i) = \sigma^2$. Define sample mean as $\bar{X} = \sum_{i=1}^n \frac{X_i}{n}$. Then,

$$\begin{aligned} Var(\bar{X}) &= Var\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} Var(\sum_{i=1}^n X_i) \\ &= \frac{1}{n^2} \sum_{i=1}^n Var(X_i) \quad (X_i \text{ are independent}) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 \\ &= \frac{\sigma^2}{n} \end{aligned}$$

So, the mean of $\bar{X}$ is same as mean of X_i but the variance decreases. This makes estimator of μ ; the result is referred as **law of large numbers**.

4.6 Conditional Expectation

Definition: The conditional expectation of Y given $X = x$ is

$$E[Y|X] = \begin{cases} \sum_y y f(y|x) & \text{discrete} \\ \int_{\mathbb{R}} y f(y|x) dy & \text{continuous} \end{cases}$$

$f(y|x)$ is conditional pmf/pdf of Y given $X = x$. Note that $E[Y|X = x]$ is a function of x .

Discrete Example:

$f_{X,Y}(x, y)$	$X = 0$	$X = 3$	$f_Y(y)$
$Y = 2$	0.11	0.34	0.45
$Y = 5$	0.00	0.05	0.05
$Y = 10$	0.29	0.21	0.50
$f_X(x)$	0.40	0.60	1

The **unconditional expectation** is

$$E[Y] = \sum_y y f_y(y) = 2 * 0.45 + 5 * 0.05 + 10 * 0.5 = 6.15$$

But conditional on $X = 3$ is

$$\begin{aligned} f(y|x=3) &= \frac{f(3, y)}{f_X(3)} \\ &= \begin{cases} \frac{34}{60} & \text{if } y = 2 \\ \frac{5}{60} & \text{if } y = 5 \\ \frac{21}{60} & \text{if } y = 10 \end{cases} \end{aligned}$$

So, the expectation conditional on $X = 3$ is

$$\begin{aligned} E[Y|X=3] &= \sum_y y f(y|3) \\ &= 2(34/60) + 5(5/60) + 10(21/60) \\ &= 5.05 \end{aligned}$$

Continuous Example:

$$f(x, y) = \frac{21}{4} x^2 y, \quad \text{if } x^2 \leq y \leq 1$$

Recall that,

$$f(y|x) = \frac{2y}{1-x^2} \quad \text{if } x^2 \leq y \leq 1$$

Thus,

$$\begin{aligned} E[Y|X] &= \int_{\mathbb{R}} y f(y|x) dy \\ &= \frac{2}{1-x^4} \int_{x^2}^1 y^2 dy \\ &= \left(\frac{2}{3}\right) \left(\frac{1-x^6}{1-x^4}\right) \end{aligned}$$

So, $E[Y|X = 0.5] = (\frac{2}{3})(\frac{1-2^6}{1-2^4}) = 0.7$

Theorem (Double Expectation):

$$E[E(Y|X)] = E[Y]$$

Remarks: The expected value (averaged over all X 's) of the conditional expected value (of $Y|X$) is the plain old expected value (of y).

Proof (continuous case): By the Unconscious Statistician,

$$\begin{aligned} E[E(Y|X)] &= \int_{\mathbb{R}} E[Y|X] f_x(x) dx \\ &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} y f(y|x) dy \right] f_x(x) dx \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} y f(y|x) f_x(x) dx dy \\ &= \int_{\mathbb{R}} y \int_{\mathbb{R}} f(x, y) dx dy \\ &= \int_{\mathbb{R}} y f_y(y) dy \\ &= E[Y] \end{aligned}$$

Example: Suppose $f(x, y) = \frac{21}{4}x^2y$, if $x^2 \leq y \leq 1$.

$$f_x(x) = \frac{21}{8}x^2(1 - x^4) \quad \text{if } -1 \leq x \leq 1$$

$$f_y(y) = \frac{7}{2}y^{\frac{5}{2}} \quad \text{if } 0 \leq y \leq 1$$

$$E[Y|X] = \frac{2}{3} \frac{1 - x^6}{1 - x^4}$$

$$\begin{aligned} E[Y] &= \int_{\mathbb{R}} y f_y(y) dy \\ &= \int_0^1 \frac{7}{2} y^{\frac{7}{2}} dy \\ &= \frac{7}{9} \end{aligned}$$

First Step Analysis

First Step Analysis

First Step method to find the mean of $Y \sim \text{Geom}(p)$. Think of Y as the number of coin flips before H appears, where $P(H) = p$. Furthermore, consider the first step of coin flip process, and

let $X = H$ or T denote the outcome of the first toss. Based on the result X of this first step.

$$\begin{aligned}
 E[Y] &= E[E(Y|X)] \\
 &= \sum_x E[Y|X]f_x(x) \\
 &= E[Y|X = T]P(X = T) + E[Y|X = H]P(X = H) \\
 &= (1 + E[Y])(1 - p) + 1(p) \quad (\text{start from scratch if } X = T)
 \end{aligned}$$

Solving, we get,

$$E[Y] = \frac{1}{p}$$

Example: Consider a sequence of coin flips. What is the expected number of flips Y until HT appears for the first time?

Clearly $Y = A + B$, where A is the number of flips until the first H appears and B is the number of subsequent flips until T appears for the first time after the sequence of H 's begins.

For instance, the sequence $TTTHHT$ corresponds to $Y = A + B = 4 + 2 = 6$.

In any case, it's obvious that A and B are iid. $Geom(p = \frac{1}{2})$ so by the previous example,

$$E[Y] = E[A] + E[B] = \frac{1}{p} + \frac{1}{p} = 4$$

Here, $p = \frac{1}{2}$ is probability of success.

Example: Again consider, a sequence of coin flips, what is the expected number of flips Y until “HH” appears for the first time?

For an instance, the sequence $TTHTTHH$ corresponds to $Y = 7$ tries,

Using an enhanced first step analysis, we see that,

$$\begin{aligned}
 E[Y] &= E[Y|T]P(T) + E[Y|H]P(H) \\
 &= E[Y|T]P(T) + E[Y|HH]P(HH|H) + E[Y|HT]P(HT|H)P(H) \\
 &= (1 + E[Y])0.5 + (2 * 0.5 + (2 + E[Y])0.5)0.5
 \end{aligned}$$

Since we have to start over once we see a T

$$= 1.5 + 0.75E[Y]$$

Solving, $E[Y] = 6$

Theorem: Expectation of sum of a random number of RVs

Suppose that $X_1, X_2, \dots$ are independent RVs all with the same mean. Also, suppose that N is a non-negative, integer-valued RV that's independent of the X_i 's. Then,

$$E[N_{i=1}^N X_i] = E[N]E[X_1]$$

Remark: In particular, note that $E[\sum_{i=1}^N X_i] \neq NE[X_1]$

Proof: By Double Expectation,

$$\begin{aligned}
E[\sum_{i=1}^N X_i] &= E[E[\sum_{i=1}^N X_i | N]] \\
&= \sum_{n=1}^{\infty} E[\sum_{i=1}^N X_i | N = n] P(N = n) \\
&= \sum_{n=1}^{\infty} (\sum_{i=1}^n X_i | N = n) P(N = n) \\
&= \sum_{n=1}^{\infty} (\sum_{i=1}^n X_i) P(N = n) \quad N \text{ and } X_i \text{'s are independent} \\
&= \sum_{n=1}^{\infty} n E[X_1] P(N = n) \\
&= E[X_1] \sum_{n=1}^{\infty} n P(N = n) \\
&= E[X_1] E[N]
\end{aligned}$$

Example: Suppose the number of times we roll a die is $N \sim \text{Pois}(10)$. If X_i denotes the value of the i^{th} toss, then the expected total of all the rolls is

$$E[\sum_{i=1}^N X_i] = E[N]E[X_1] = 10 * 3.5 = 35$$

Theorem: Under the same conditions as before

$$\text{Var}(\sum_{i=1}^N X_i) = E[N]\text{Var}(X_1) + (E[X_1])^2 \text{Var}(N)$$

4.6.1 Standard Conditioning Argument

Theorem/Proof (Computing Probabilities by Conditioning)

Let A be some event, and define the RV Y as the following indicator function:

$$Y = I_A = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Then,

$$\begin{aligned}
E[Y] &= \sum_y y f_y(y) \\
&= P(Y = 1) \\
&= P(A)
\end{aligned}$$

Similarly, for any RV X , we have,

$$\begin{aligned}
E[Y/X = x] &= \sum_y y f(y/x) \\
&= P(Y = 1/X = x) \\
&= P(A/X = x)
\end{aligned}$$

Theorem: If X is a continuous RV (similar result if X is discrete) then,

$$P(A) = \int_{\mathbb{R}} P(A/X = x) f_x(x) dx$$

Proof:

$$\begin{aligned} P(A) &= E[Y] \\ &= E[E[Y/X]] \quad (\text{double expectation}) \\ &= \int_{\mathbb{R}} E[Y/X] f_x(x) dx \quad (\text{LOTUS}) \\ &= \int_{\mathbb{R}} P(A/X = x) f_x(x) dx \quad (\text{Since, } Y = 1_A) \end{aligned}$$

Theorem: If X and Y are independent continuous RV, with pdf $f_X(\cdot)$ and cdf $F_X(\cdot)$ respectively. Then,

$$P(Y \leq X) = \int_{\mathbb{R}} F_Y(x) f_X(x) dx$$

where, $F_Y(x)$ is CDF of Y at x and $f_X(x)$ is PDF of X at x .

Proof: Let the event $A = \{Y \leq X\}$

$$\begin{aligned} P(Y \leq X) &= \int_{\mathbb{R}} P(Y \leq X/X = x) f_X(x) dx \\ &= \int_{\mathbb{R}} P(Y \leq X/X = x) f_X(x) dx \quad (X \text{ and } Y \text{ are independent because } Y \leq X \text{ doesn't depend on } X) \\ &= \int_{\mathbb{R}} F_Y(x) f_X(x) dx \end{aligned}$$

Example: If $X \sim \text{Exp}(\alpha)$ and $Y \sim \text{Exp}(\beta)$ are independent RVs, then,

$$\begin{aligned} P(Y \leq X) &= \int_{\mathbb{R}} F_Y(x) f_X(x) dx \\ &= \int_0^{\infty} (1 - e^{-\beta x}) \alpha e^{-\alpha x} dx \\ &= \frac{\beta}{\alpha + \beta} \end{aligned}$$